Python Variables, Data Types, and Naming Conventions

Variables

A variable is a name used to store data in memory so that it can be used later in a program.

```
Example:
name = 'Rajeev'
age = 25
```

Data Types in Python

1. int (Integer) - Whole numbers, e.g., age = 21

2. float - Decimal numbers, e.g., price = 99.99

3. str (String) - Text in quotes, e.g., name = 'Rajeev'

4. bool (Boolean) - True or False

5. complex - Complex numbers, e.g., z = 3 + 4j

Checking Data Type

Use type() function.
print(type(age))

Variable Naming Rules

• Cannot start with a number
• No spaces allowed
• No special symbols except underscore (_)
• Cannot use Python keywords

Naming Conventions

Use snake_case for variables: student_name, total_marks

Use UPPER_CASE for constants: PI, MAX_USERS

Practice Program

```
name = input("Enter your name: ")
age = int(input("Enter your age: "))
height = float(input("Enter your height: "))

print(type(name))
print(type(age))
print(type(height))
```